

Exam 2020 - sample solutions

High-Frequency Trading and Stock Market Volatility

This sample solution is one of many possible solutions. The grade for the exam will not be based on how closely you follow a particular suggested solution but rather on the overall quality of your empirical analysis.

Abstract

High-frequency trading increases trading efficiency due to advanced technology but might also increase stock volatility. We empirically investigate this using the staggered adoption of the 2006 NYSE Hybrid Market transition. We find a positive effect of the rollover to the new system on trading activity. Using a 2-stage IV approach, we also find an increase in the high-minus-low range but no significant effect on stock return volatility.

1 Introduction

In this study, we examine the effect of high-frequency trading on stock volatility. While high-frequency trading increases trading efficiency due to advanced technology, it might also increase stock volatility as possibly "much more orders (quotes) relative to the actual number of executed trades" might be submitted.

To investigate the impact of high-frequency trading on stock volatility we use the staggered adoption of the 2006 NYSE Hybrid Market transition. The staggered nature allows us to differentiate between stocks that have already rolled over and stocks that are yet to roll over.

We first investigate whether rollover to the NYSE Hybrid Market affects high-frequency trading activity. We find a positive effect of the rollover to the new system on trading activity.

Next, we investigate the main research question: Does high-frequency trading affect stock volatility? To address endogeneity concerns, we use the 2-stage IV approach where we instrument high-frequency trading by the transition to the new system. The result depends on our proxy for stock volatility: we find an increase in the spread but no effect on stock return volatility.

2 Data

We employ 3 datasets: one that includes the event dates, one with trading proxies and a third one with CRSP variables. Trading proxies and CRSP variables are on daily basis and cover the years 2006 and 2007. We used the unique identifier "TICKER" to merge all 3 datasets.

We clean the data in the following ways. First, we keep only days with non-zero trading volume (excluding, for instance, holidays). We also remove missing observations, and we exclude stocks that trade less than 200 days in our sample period and that have a price below USD 5 or above USD 1000.

To proxy for volatility, we first use the high-minus-low range as defined in Parkinson (1980). Second, we calculate the rolling standard deviation of returns. In later regressions, we use both proxies.

As control variables, we use market capitalization, (proportional quoted) bid-ask spread, annualized turnover and price inverse. We also winsorize all variables at the 1% (99%) level.

Table 1 displays the summary statistics (including mean, median and standard deviation) of the most important variables. Importantly, the difference between our 2 volatility proxies is not large, with "RANGE" having a lower mean and standard deviation.

3 Empirical analysis

We first investigate whether rollover to the NYSE Hybrid Market affects high-frequency trading activity. We do this by regressing the number of quote updates to the dollar volume executed per USD ("QTE_DVOL") on the dummy variable "post" equal to 1 after the introduction of the new system, and 0 otherwise. By running a panel regression with twoway fixed effects, we essentially achieve a difference-in-difference approach where we compare stocks that are not yet rolled over to the new system to stocks that are already rolled over.

Table 2 presents the results of this analysis, with Column (1) only showing the regression without additional controls and Column (2) with additional controls. In both columns, the coefficient is statistically significant and positive, indicating a positive effect of the rollover to the new system on trading activity.

Next, we investigate the main research question: Does high-frequency trading affect stock volatility? A direct regression of volatility on trading volume suffers from endogeneity due to possible reverse causality (as trading volume is a decision that depends on volatility) or omission of explanatory variables (as other factors might also affect volatility). To address endogeneity, we use the 2-stage IV approach where we instrument high-frequency trading by the transition to the new system (using "post" as instrument).

Table 3 presents the results of both the first stage (Columns (1) and (2)) and the second stage (Columns (3) and (4)) using the volatility proxies SDRET (Columns (1) and (3)) and RANGE (Columns (2) and (4)), respectively. The results of the second stage can be interpreted as following:

When using SDRET as dependent variable, there is no significant effect of high-frequency trading on stock volatility, even though the coefficient is negative (-0.013). However, when using RANGE, there is a positive and significant effect of high-frequency trading on stock volatility. The difference in the interpretation might depend on the inherent differences of the two proxies: high-frequency trading might increase the high-minus-low range but has no significant effect on the standard deviation of returns.

4 Conclusion

We examine the effect of high-frequency trading on stock volatility. High-frequency trading increases trading efficiency due to advanced technology but might also increase stock volatility.

To investigate the impact of high-frequency trading on stock volatility we use the staggered adoption of the 2006 NYSE Hybrid Market transition. The staggered nature allows us to differentiate between stocks that have already rolled over and stocks that are yet to roll over. We find a positive effect of the rollover to the new system on trading activity.

We examine the effect of high-frequency trading on stock volatility using a 2-stage IV approach where we instrument high-frequency trading by the transition to the new system. We find an increase in the spread but no effect on stock return volatility.

5 References

Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of business*, 61-65.

6 Appendix

Table 1: **Summary statistics of the most important variables used in this study**

This table shows the summary statistics for the variables used in this study. Detailed variable definitions can be found in the R-code or the assignment description. Variables have been winsorized at the 1% (99%) level.

	n	mean	median	sd	min	max
QTE_DVOL	285651.00	6.93	4.90	6.33	0.58	38.05
RANGE	285651.00	1.37	1.16	0.81	0.30	5.29
SDRET	285647.00	1.50	1.28	0.93	0.23	5.98
bidask	285651.00	0.00	0.00	0.00	0.00	0.01
mcap	285651.00	1.16	0.99	1.36	-1.54	5.29
inv_prc	285651.00	0.04	0.03	0.03	0.01	0.17
turn	285651.00	2206.73	1576.89	2060.75	202.56	14758.25

Table 2: **Effect of the Hybrid Market transition on trading activity**

This table shows the results of the OLS regression of the proxy for trading activity (QTE_DVOL) on the dummy variable "post" which is equal to 1 for period after the Hybrid Market transition, and 0 otherwise, and additional controls (only Column (2)). Detailed variable definitions can be found in the R-code or the assignment description. Variables have been winsorized at the 1% (99%) level. Standard errors are robust.

	<i>Dependent variable:</i>	
	QTE_DVOL	
	(1)	(2)
post	1.190*** t = 11.932	1.254*** t = 14.061
bidask		66.614*** t = 7.445
mcap		-2.187*** t = -7.364
inv_prc		78.398*** t = 8.590
turn		-0.001*** t = -30.999
Observations	285651	285651
Model	OLS	OLS
Stock FE	yes	yes
Day FE	yes	yes
Cluster SE	by stock	by stock
F-stat: post=0	142.36	197.73
Adjusted R ²	-0.002	0.168
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Table 3: **Effect of the Hybrid Market transition on stock volatility**

This table shows the results of both the first stage (Columns (1) and (2)) and the second stage (Columns (3) and (4)) of a 2-stage IV regression using the volatility proxies SDRET (Columns (1) and (3)) and RANGE (Columns (2) and (4)), respectively. Detailed variable definitions can be found in the R-code or the assignment description. Variables have been winsorized at the 1% (99%) level. Standard errors are robust.

	<i>Dependent variable:</i>			
	SDRET	RANGE	SDRET	RANGE
	(1)	(2)	(3)	(4)
QTE_DVOL	0.001** t = 2.168	0.004*** t = 4.211	-0.013 t = -1.512	0.035*** t = 4.859
bidask	3.300** t = 2.283	15.362*** t = 10.680	4.310*** t = 2.673	13.151*** t = 8.995
mcap	0.036 t = 1.102	0.190*** t = 3.602	0.005 t = 0.139	0.258*** t = 4.564
inv_prc	1.127 t = 1.595	3.957*** t = 3.758	2.257** t = 2.432	1.484 t = 1.163
turn	0.0001*** t = 43.088	0.0002*** t = 62.394	0.0001*** t = 12.157	0.0002*** t = 31.215
Observations	285647	285651	285647	285651
Model	OLS	OLS	2SLS	2SLS
Stock FE	yes	yes	yes	yes
Day FE	yes	yes	yes	yes
Cluster SE	by stock	by stock	by stock	by stock
Adjusted R ²	0.037	0.207	0.034	0.185

Note:

*p<0.1; **p<0.05; ***p<0.01